

03 - Direction fields end; Existence and Uniqueness

1. **Daily Problem.** Draw the direction field for this ODE, with a few representative solutions: $\frac{dy}{dt} = -\frac{t}{y}$.

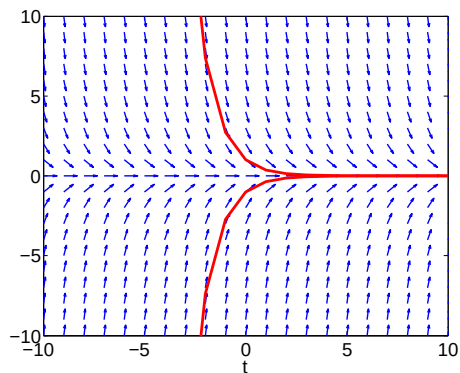
2. What do the terms “exists” and “unique” mean to you? What do these terms mean in the context of solutions of algebraic equations such as $x + 3 = 5$, $x^2 = 4$, or $e^x = -3$?

3. In the context of solving an initial value problem:

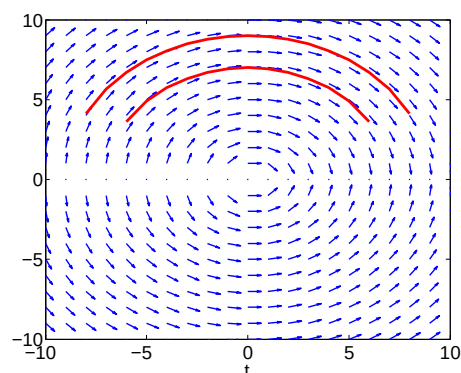
- “Existence” refers to whether there is a solution that exists, meaning whether there is a function that satisfies the IVP (solves the ODE and satisfies the IC). Note this function might only solve the IVP on a limited interval.
- “Unique” or “uniqueness” refers to whether a single function satisfies the IVP, or more than one.

So when do solutions of IVPs exist? When are they unique? Looking at these direction fields, from worksheet 2 #7, what can you say about existence and uniqueness of solutions in each case?

(b) $\frac{dy}{dt} = -y$.



(d) $\frac{dy}{dt} = -\frac{t}{y}$.



The Existence and Uniqueness Theorem For First-Order Non Linear ODEs.

Consider the IVP

$$\frac{dy}{dt} = f(t,y), \quad y(t_0) = y_0.$$

Assume the functions f and $\frac{\partial f}{\partial y}$ are both continuous on a rectangle that contains the IC point (t_0, y_0) :

$$R = \{(t,y) : t \in (\alpha, \beta), y \in (a,b)\}, \quad (t_0, y_0) \in R.$$

Then there exists an interval $I = (t_0 - h, t_0 + h)$ contained in (α, β) such that the IVP has a solution $y(t)$ defined on the interval I (hence the solution exists), and that solution is unique on the interval I .

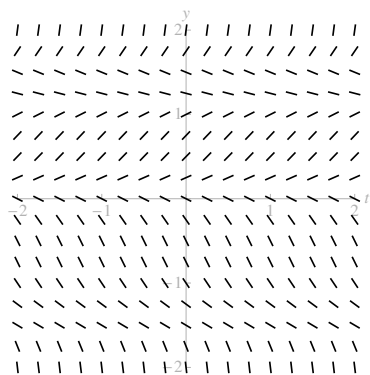
4. Consider the ODE $y' = 2\sqrt{y}$. Do a quick slope field for this ODE and draw a few representative solutions. Does the Existence and Uniqueness Theorem For First-Order Non Linear ODEs apply to this ODE? Use your work to give you an idea of where two distinct solutions to the same IVP might “branch off”, and find those two solutions.

5. Consider the differential equation $x' = 3x + t$. Using a few isoclines (including all nullclines), sketch by hand the direction field of this differential equation. Then, on your direction field, sketch the solution curves corresponding to these initial conditions: $x(-1) = 0$ and $x(0) = 0$. Consider the interval $[-5, 5]$ for both t and x .

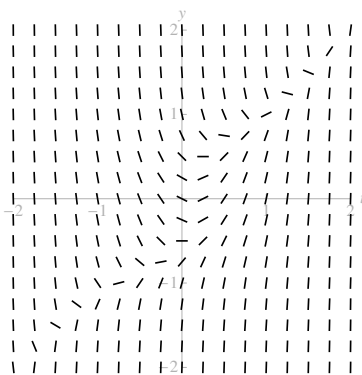
6. Sketch a direction field for the ODE $y' = y^2 - t$. Sketch the solution with $y(1) = 1/2$ on your direction field. Pay particular attention to the long-term behaviour of that solution.

7. Daily Problem. Please try this question in preparation for next class.

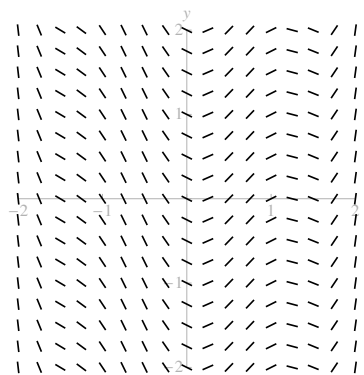
Here are three direction fields.



(I)



(II)



(III)

- (a) For each of the following ODEs, find the picture above which shows its direction field.

i. $\frac{dy}{dt} = t^5 - 4t^3 + 4t - \frac{1}{2}$.

ii. $\frac{dy}{dt} = y^5 - 4y^3 + 4y - \frac{1}{2}$.

- (b) For each ODE, can you also draw a few representative solutions? Are some solutions very similar to others, and if so, how? Do some of the solutions cross?